THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 17

Medallion Architecture — Bronze, Silver, Gold

SECTION 01

What Medallion Architecture Is

CORE CONCEPT

The medallion architecture describes a series of data layers that denote the quality of data stored in the lakehouse. Its goal is to incrementally and progressively improve the structure and quality of data as it flows through each layer, from Bronze to Silver to Gold. Databricks recommends this multi-layered approach for building a single source of truth for enterprise data products.

This architecture guarantees atomicity, consistency, isolation, and durability as data passes through multiple layers of validations and transformations before being stored in a layout optimized for efficient analytics. Each layer can contain one or more tables, and a domain can own its Bronze, Silver, and Gold tables end to end.

SECTION 02

Bronze — Raw, As-Is Data

Bronze is the landing zone and the historical record. Data lands exactly as received from the source system, typically append-only, with technical metadata added at ingestion time, such as load timestamp, source file name, and batch identifier.

- Never transform, never drop fields; Bronze absorbs upstream chaos as-is.
- Written by ingestion tools such as Auto Loader or Lakeflow Connect.
- Read only by Silver pipelines, not consumed directly by BI tools or analysts.

KEY RULE

Bronze must be append-only. If Bronze is mutable, the replay guarantee that justifies the whole architecture disappears; you lose the ability to rebuild everything downstream when transformation logic changes, without needing to re-ingest from the original source.

SECTION 03

Silver — Cleaned and Conformed Data

Silver is where raw chaos becomes a single, trusted, agreed-upon version of the data. Types are fixed, invalid records are filtered or flagged, duplicates are removed, and tables are joined into a more usable, conformed structure.

- Schema is enforced (schema-on-write), unlike Bronze's schema-on-read approach.
- Deduplication and data quality expectations are applied at this boundary.
- Read by Gold pipelines, data scientists, and ML teams, not typically by end-user dashboards directly.

COMMON MISTAKE

Treating Silver as just a staging area with no rules is one of the two mistakes that most often sink medallion implementations in practice. Silver is a contract, not a suggestion: it enforces schema and

quality rules, and downstream consumers rely on that guarantee.

SECTION 04

Gold — Business-Ready, Aggregated Data

Gold holds aggregated, use-case-optimized tables, tailored to specific business needs such as dashboards, reports, or machine learning feature tables. This is typically the layer BI tools query directly.

- One table per clearly defined use case is the recommended pattern.
- Read by BI tools, analysts, dashboards, and ML serving.

COMMON MISTAKE

Materializing too many Gold tables until nobody knows which one is correct is the second common failure pattern. Gold should serve specific, well-understood consumers, not become a dumping ground of loosely related aggregates.

SECTION 05

The Three Layers at a Glance

Layer	Data State	Who Writes	Who Reads
Bronze	Raw, unmodified	Ingestion tools (Auto Loader, Lakeflow Connect)	Silver pipelines only
Silver	Cleaned, validated, deduplicated	Silver pipelines	Gold pipelines, data scientists, ML teams
Gold	Aggregated, use-case optimized	Gold pipelines	BI tools, analysts, dashboards, ML serving

SECTION 06

How This Maps to Databricks Tools

On Databricks specifically, several platform pieces work together to implement this pattern.

- Unity Catalog provides the catalog.schema.table namespace; layers can live at the catalog level or the schema level, both are valid layouts.
- Auto Loader handles incremental file ingestion into Bronze.
- Delta Lake MERGE or streaming tables power the Bronze-to-Silver and Silver-to-Gold hops.
- Lakeflow Declarative Pipelines is Databricks' recommended way to build Bronze, Silver, and Gold tables using streaming tables and materialized views that refresh incrementally.

```
-- Example: schema-per-layer setup using Unity Catalog
CREATE CATALOG IF NOT EXISTS prod;

CREATE SCHEMA IF NOT EXISTS prod.bronze;
CREATE SCHEMA IF NOT EXISTS prod.silver;
CREATE SCHEMA IF NOT EXISTS prod.gold;

GRANT USE SCHEMA ON SCHEMA prod.bronze TO `data-engineers`;
GRANT USE SCHEMA ON SCHEMA prod.silver TO `data-engineers`;
GRANT SELECT ON SCHEMA prod.gold TO `data-analysts`;
```

SECTION 07

Data Flow Through the Layers

```
  SOURCE SYSTEMS
        |
        v
  BRONZE  (raw, append-only, as-is)
        |  clean, validate, dedupe, enforce schema
        v
  SILVER  (trusted, conformed, single source of truth)
        |  aggregate, shape for specific use cases
        v
  GOLD    (business-ready, one table per use case)
        |
        v
  BI TOOLS / DASHBOARDS / ML SERVING
```

REMEMBER THIS FOR INTERVIEWS

Each hop through Bronze, Silver, and Gold adds validation, transformation, and business logic. If something breaks in Gold, you trace back through Silver to Bronze; because raw data is never lost, schema changes or business logic updates can be replayed from Bronze without re-ingesting from the original source.

SECTION 08

Glossary — Day 17 Terms

Term	Definition
Medallion Architecture	A data design pattern organizing a lakehouse into Bronze, Silver, and Gold layers of progressively improving data quality.
Bronze Layer	The raw, append-only landing zone preserving source data exactly as received, plus ingestion metadata.

Silver Layer	The cleaned, deduplicated, schema-enforced layer representing a single agreed-upon version of the data.
Gold Layer	The aggregated, business-level layer optimized for specific consumers like dashboards and ML serving.
Lakeflow Declarative Pipelines	Databricks' recommended framework for building medallion layers using streaming tables and incrementally refreshing materialized views.
Schema-on-Read vs Schema-on-Write	Bronze typically uses schema-on-read (flexible, as-is); Silver enforces schema-on-write (strict, validated) at ingestion.

SECTION 09

Interview Questions & Answers

Q1. What is the medallion architecture?

A. A data design pattern that organizes a lakehouse into Bronze, Silver, and Gold layers, incrementally and progressively improving the structure and quality of data as it flows through each layer.

Q2. Why must the Bronze layer be append-only?

A. Because Bronze serves as the historical record and replay source. If Bronze is mutable, the ability to rebuild all downstream layers from scratch when transformation logic changes is lost, without needing to re-ingest from the original source system.

Q3. What is the difference between what Silver and Gold each contain?

A. Silver holds cleaned, validated, deduplicated, schema-enforced data representing a single trusted version. Gold holds aggregated, use-case-optimized tables built specifically for consumers like dashboards, reports, or ML serving.

Q4. What is a common mistake teams make with the Silver layer?

A. Treating Silver as just a staging area with no rules, rather than enforcing schema and data quality expectations at that boundary, which undermines the reliability guarantee the rest of the architecture depends on.

Q5. What is a common mistake teams make with the Gold layer?

A. Materializing too many Gold tables until nobody knows which one is correct, instead of keeping Gold scoped to one clearly defined table per specific use case.

Q6. What Databricks tools are typically used to implement medallion architecture?

A. Unity Catalog provides the catalog.schema.table namespace for organizing layers, Auto Loader handles incremental ingestion into Bronze, and Lakeflow Declarative Pipelines is the recommended way to build the Bronze, Silver, and Gold hops using streaming tables and materialized views.

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